

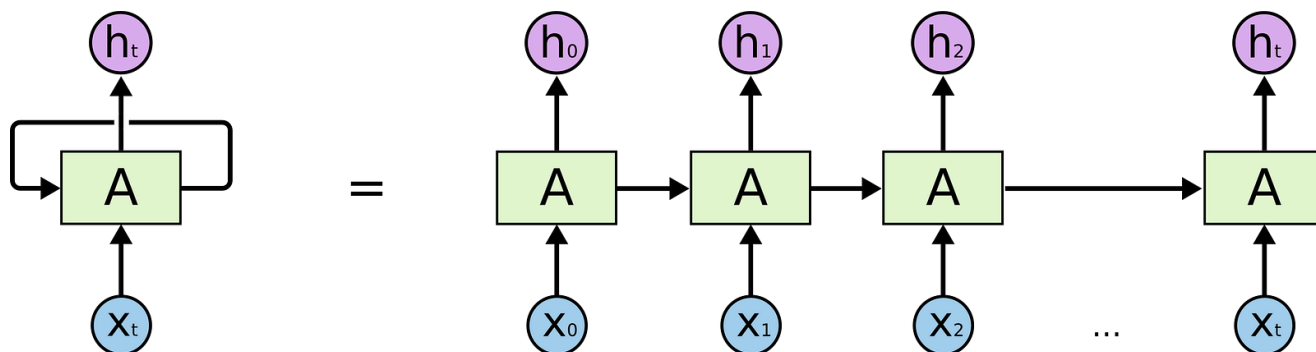
Things to remember

- If you are not familiar with keras or neural networks, refer to this kernel/tutorial of mine: <https://www.kaggle.com/thebrownviking20/intro-to-keras-with-breast-cancer-data-ann>
- Your doubts and curiosity about time series can be taken care of here: <https://www.kaggle.com/thebrownviking20/everything-you-can-do-with-a-time-series>
- Don't let the explanations intimidate you. It's simpler than you think.
- Eventually, I will add more applications of LSTMs. So stay tuned for more!
- The code is inspired from Kirill Eremenko's Deep Learning Course: <https://www.udemy.com/deeplearning/>

✓ Recurrent Neural Networks

In a recurrent neural network we store the output activations from one or more of the layers of the network. Often these are hidden layer activations. Then, the next time we feed an input example to the network, we include the previously-stored outputs as additional inputs. You can think of the additional inputs as being concatenated to the end of the "normal" inputs to the previous layer. For example, if a hidden layer had 10 regular input nodes and 128 hidden nodes in the layer, then it would actually have 138 total inputs (assuming you are feeding the layer's outputs into itself à la Elman) rather than into another layer). Of course, the very first time you try to compute the output of the network you'll need to fill in those extra 128 inputs with 0s or something.

Source: [Quora](#)



Source: [Medium](#)

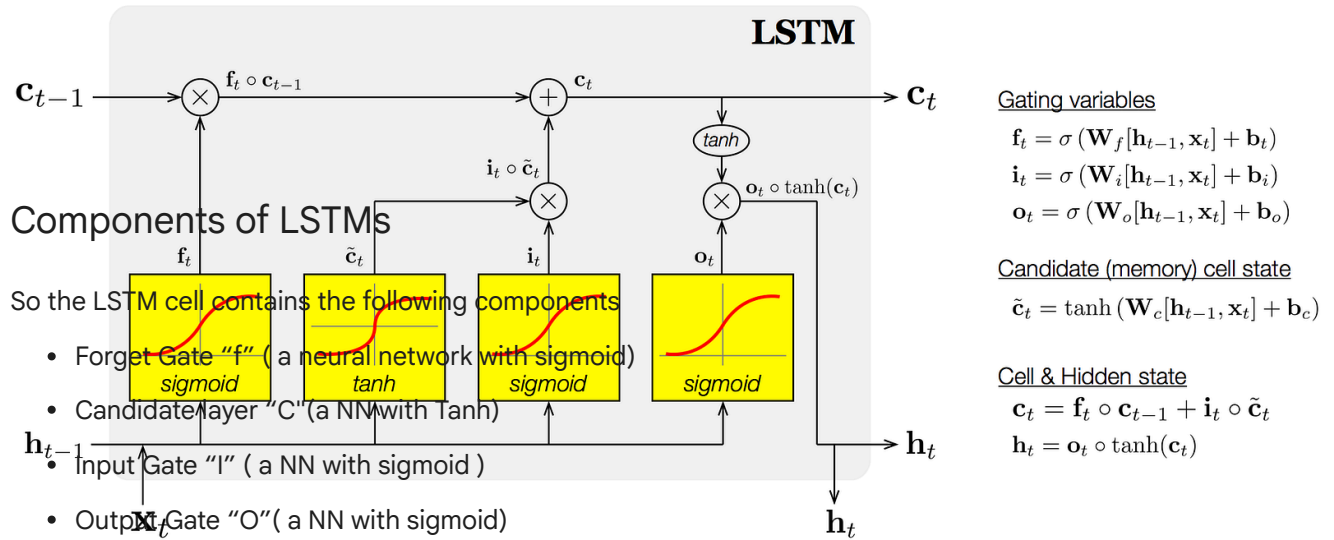
Let me give you the best explanation of Recurrent Neural Networks that I found on internet:

<https://www.youtube.com/watch?v=UNmqTiOnRfg&t=3s>

Now, even though RNNs are quite powerful, they suffer from ****Vanishing gradient problem**** which hinders them from using long term information, like they are good for storing memory 3-4 instances of past iterations but larger number of instances don't provide good results so we don't just use regular RNNs. Instead, we use a better variation of RNNs: **Long Short Term Networks(LSTM)**.

What is Vanishing Gradient problem?

Vanishing gradient problem is a difficulty found in training artificial neural networks with gradient-based learning methods and backpropagation. In such methods, each of the neural network's weights receives an update proportional to the partial derivative of the error function with respect to the current weight in each iteration of training. The problem is that in some cases, the gradient will be vanishingly small, effectively preventing the weight from changing its value. In the worst case, this may completely stop the neural network from further training. As one example of the problem cause, traditional activation functions such as the hyperbolic tangent function have gradients in the range (0, 1), and backpropagation computes gradients by the chain rule. This has the effect of multiplying n of these small numbers to compute gradients of the "front" layers in an n-layer



Source: [Medium](https://medium.com/deep-math-machine-learning-ai/chapter-10-1-deep-input-to-the-lstm-cell-at-any-step-are-x-current-input-2147769105)

The best LSTM explanation on internet: <https://medium.com/deep-math-machine-learning-ai/chapter-10-1-deep-input-to-the-lstm-cell-at-any-step-are-x-current-input-2147769105>

Working of gates in LSTMs

First, LSTM cell takes the previous memory state C_{t-1} and does element wise multiplication with forget gate (f) to decide if present memory state C_t . If forget gate value is 0 then previous memory state is completely forgotten else f forget gate value is 1 then previous memory state is completely passed to the cell (Remember f gate gives values between 0 and 1).

$$C_t = C_{t-1} * f_t$$

Calculating the new memory state:

$$C_t = C_t + (I_t * C_t)$$

Now, we calculate the output:

$$H_t = \tanh(C_t)$$

And now we get to the code...

I will use LSTMs for predicting the price of stocks of IBM for the year 2017

```
# Importing the libraries
import numpy as np
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from keras.models import Sequential
from keras.layers import Dense, LSTM, Dropout, GRU, Bidirectional
from keras.optimizers import SGD
import math
from sklearn.metrics import mean_squared_error
```

```
# Some functions to help out with
def plot_predictions(test, predicted):
```

```
plt.plot(test, color='red',label='Real IBM Stock Price')
plt.plot(predicted, color='blue',label='Predicted IBM Stock Price')
plt.title('IBM Stock Price Prediction')
plt.xlabel('Time')
plt.ylabel('IBM Stock Price')
plt.legend()
plt.show()
```

```
def return_rmse(test,predicted):
    rmse = math.sqrt(mean_squared_error(test, predicted))
    print("The root mean squared error is {}".format(rmse))
```

```
# First, we get the data
dataset = pd.read_csv('IBM_2006-01-01_to_2018-01-01.csv', index_col='Date', parse_dates=['Date'])
dataset.head()
```

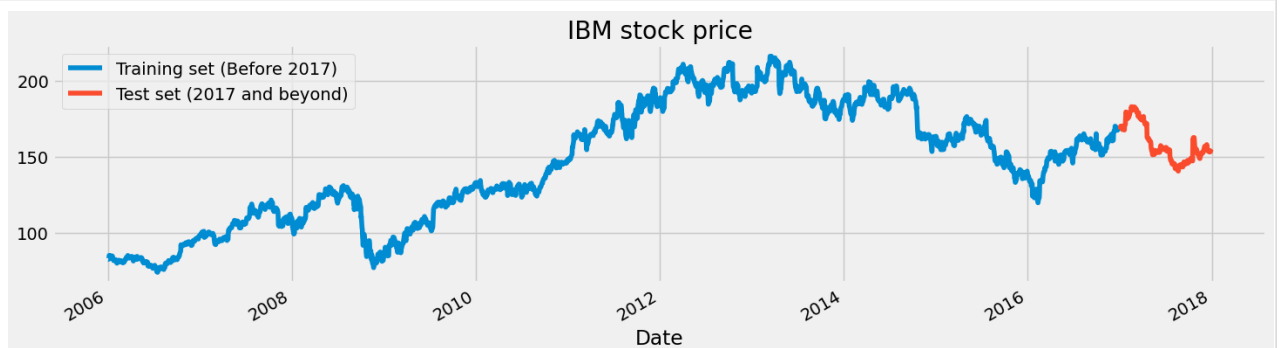
	Open	High	Low	Close	Volume	Name	
Date							
2006-01-03	82.45	82.55	80.81	82.06	11715200	IBM	
2006-01-04	82.20	82.50	81.33	81.95	9840600	IBM	
2006-01-05	81.40	82.90	81.00	82.50	7213500	IBM	
2006-01-06	83.95	85.03	83.41	84.95	8197400	IBM	
2006-01-09	84.10	84.25	83.38	83.73	6858200	IBM	

Next steps: [Generate code with dataset](#) [View recommended plots](#) [New interactive sheet](#)

```
# Checking for missing values
training_set = dataset[:'2016'].iloc[:,1:2].values
test_set = dataset['2017:'].iloc[:,1:2].values
```

Start coding or [generate](#) with AI.

```
# We have chosen 'High' attribute for prices. Let's see what it looks like
dataset["High"][:'2016'].plot(figsize=(16,4),legend=True)
dataset["High"]['2017:'].plot(figsize=(16,4),legend=True)
plt.legend(['Training set (Before 2017)','Test set (2017 and beyond)'])
plt.title('IBM stock price')
plt.show()
```



```
# Scaling the training set
sc = MinMaxScaler(feature_range=(0,1))
training_set_scaled = sc.fit_transform(training_set)
```

```
# Since LSTMs store long term memory state, we create a data structure with 60 timesteps and 1 output
# So for each element of training set, we have 60 previous training set elements
X_train = []
```

```
y_train = []
for i in range(60,2769):
    X_train.append(training_set_scaled[i-60:i,0])
    y_train.append(training_set_scaled[i,0])
X_train, y_train = np.array(X_train), np.array(y_train)
```

```
# Reshaping X_train for efficient modelling
X_train = np.reshape(X_train, (X_train.shape[0],X_train.shape[1],1))
```

```
# The LSTM architecture
regressor = Sequential()
# First LSTM layer with Dropout regularisation
regressor.add(LSTM(units=50, return_sequences=True, input_shape=(X_train.shape[1],1)))
regressor.add(Dropout(0.2))
# Second LSTM layer
regressor.add(LSTM(units=50, return_sequences=True))
regressor.add(Dropout(0.2))
# Third LSTM layer
regressor.add(LSTM(units=50, return_sequences=True))
regressor.add(Dropout(0.2))
# Fourth LSTM layer
regressor.add(LSTM(units=50))
regressor.add(Dropout(0.2))
# The output layer
regressor.add(Dense(units=1))

# Compiling the RNN
regressor.compile(optimizer='rmsprop',loss='mean_squared_error')
# Fitting to the training set
regressor.fit(X_train,y_train,epochs=50,batch_size=32)
```

```

85/85 ————— 11s 118ms/step - loss: 0.0017
Epoch 43/50
85/85 ————— 10s 118ms/step - loss: 0.0017
Epoch 44/50
85/85 ————— 10s 117ms/step - loss: 0.0017
Epoch 45/50
85/85 ————— 9s 104ms/step - loss: 0.0016
Epoch 46/50
85/85 ————— 12s 120ms/step - loss: 0.0017
Epoch 47/50
85/85 ————— 10s 118ms/step - loss: 0.0017
Epoch 48/50
85/85 ————— 10s 118ms/step - loss: 0.0016
Epoch 49/50
85/85 ————— 10s 119ms/step - loss: 0.0015
Epoch 50/50
85/85 ————— 9s 108ms/step - loss: 0.0016
<keras.src.callbacks.history.History at 0x7afc95b9daf0>

```

```

# Now to get the test set ready in a similar way as the training set.
# The following has been done so first 60 entires of test set have 60 previous values which is impos
# 'High' attribute data for processing
dataset_total = pd.concat((dataset["High"][:'2016'],dataset["High"]['2017':]),axis=0)
inputs = dataset_total[len(dataset_total)-len(test_set) - 60:].values
inputs = inputs.reshape(-1,1)
inputs = sc.transform(inputs)

```

```

# Preparing X_test and predicting the prices
X_test = []
for i in range(60,311):
    X_test.append(inputs[i-60:i,0])
X_test = np.array(X_test)
X_test = np.reshape(X_test, (X_test.shape[0],X_test.shape[1],1))
predicted_stock_price = regressor.predict(X_test)
predicted_stock_price = sc.inverse_transform(predicted_stock_price)

```

```

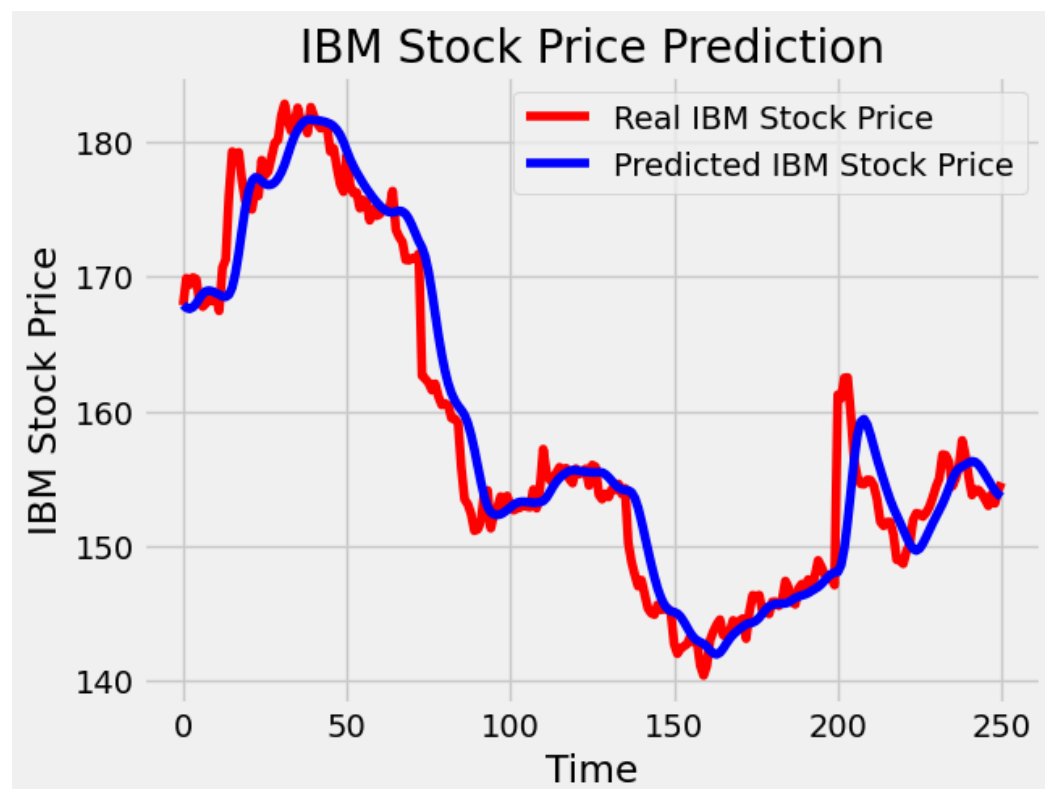
8/8 ————— 2s 198ms/step

```

```

# Visualizing the results for LSTM
plot_predictions(test_set,predicted_stock_price)

```



```
# Evaluating our model
return_rmse(test_set,predicted_stock_price)
```

The root mean squared error is 3.02497018913299.

Truth be told. That's one awesome score.

LSTM is not the only kind of unit that has taken the world of Deep Learning by a storm. We have **Gated Recurrent Units(GRU)**. It's not known, which is better: GRU or LSTM because they have comparable performances. GRUs are easier to train than LSTMs.

✓ Gated Recurrent Units

In simple words, the GRU unit does not have to use a memory unit to control the flow of information like the LSTM unit. It can directly makes use of the all hidden states without any control. GRUs have fewer parameters and thus may train a bit faster or need less data to generalize. But, with large data, the LSTMs with higher expressiveness may lead to better results.

They are almost similar to LSTMs except that they have two gates: reset gate and update gate. Reset gate determines how to combine new input to previous memory and update gate determines how much of the previous state to keep. Update gate in GRU is what input gate and forget gate were in LSTM. We don't have the second non linearity in GRU before calculating the output, neither they have the output gate.

Source: [Quora](#)



```
# The GRU architecture
regressorGRU = Sequential()
# First GRU layer with Dropout regularisation
regressorGRU.add(GRU(units=50, return_sequences=True, input_shape=(X_train.shape[1],1), activation='tanh'))
regressorGRU.add(Dropout(0.2))
# Second GRU layer
regressorGRU.add(GRU(units=50, return_sequences=True, input_shape=(X_train.shape[1],1), activation='tanh'))
regressorGRU.add(Dropout(0.2))
# Third GRU layer
regressorGRU.add(GRU(units=50, return_sequences=True, input_shape=(X_train.shape[1],1), activation='tanh'))
regressorGRU.add(Dropout(0.2))
# Fourth GRU layer
regressorGRU.add(GRU(units=50, activation='tanh'))
regressorGRU.add(Dropout(0.2))
# The output layer
regressorGRU.add(Dense(units=1))
# Compiling the RNN
regressorGRU.compile(optimizer=SGD(learning_rate=0.01, momentum=0.9, nesterov=False), loss='mean_squared_error')
# Fitting to the training set
regressorGRU.fit(X_train,y_train,epochs=50,batch_size=150)
```

```

19/19 ————— 5s 274ms/step - loss: 0.0024
Epoch 32/50
19/19 ————— 11s 288ms/step - loss: 0.0025
Epoch 33/50
19/19 ————— 11s 348ms/step - loss: 0.0023
Epoch 34/50
19/19 ————— 5s 287ms/step - loss: 0.0021
Epoch 35/50
19/19 ————— 7s 344ms/step - loss: 0.0022
Epoch 36/50
19/19 ————— 10s 312ms/step - loss: 0.0022
Epoch 37/50
19/19 ————— 10s 280ms/step - loss: 0.0023
Epoch 38/50
19/19 ————— 10s 285ms/step - loss: 0.0022
Epoch 39/50
19/19 ————— 10s 287ms/step - loss: 0.0022
Epoch 40/50
19/19 ————— 6s 335ms/step - loss: 0.0022
Epoch 41/50
19/19 ————— 6s 280ms/step - loss: 0.0021
Epoch 42/50
19/19 ————— 5s 282ms/step - loss: 0.0022
Epoch 43/50
19/19 ————— 10s 275ms/step - loss: 0.0023
Epoch 44/50
19/19 ————— 10s 271ms/step - loss: 0.0020
Epoch 45/50
19/19 ————— 10s 273ms/step - loss: 0.0021
Epoch 46/50
19/19 ————— 11s 294ms/step - loss: 0.0022
Epoch 47/50
19/19 ————— 11s 347ms/step - loss: 0.0021
Epoch 48/50
19/19 ————— 5s 273ms/step - loss: 0.0020
Epoch 49/50
19/19 ————— 10s 272ms/step - loss: 0.0020
Epoch 50/50
19/19 ————— 11s 278ms/step - loss: 0.0022
<keras.src.callbacks.history.History at 0x7afc88fda7e0>

```

The current version uses a dense GRU network with 100 units as opposed to the GRU network with 50 units in previous version

```

# Preparing X_test and predicting the prices
X_test = []
for i in range(60,311):
    X_test.append(inputs[i-60:i,0])
X_test = np.array(X_test)
X_test = np.reshape(X_test, (X_test.shape[0],X_test.shape[1],1))
GRU_predicted_stock_price = regressorGRU.predict(X_test)
GRU_predicted_stock_price = sc.inverse_transform(GRU_predicted_stock_price)

```

```

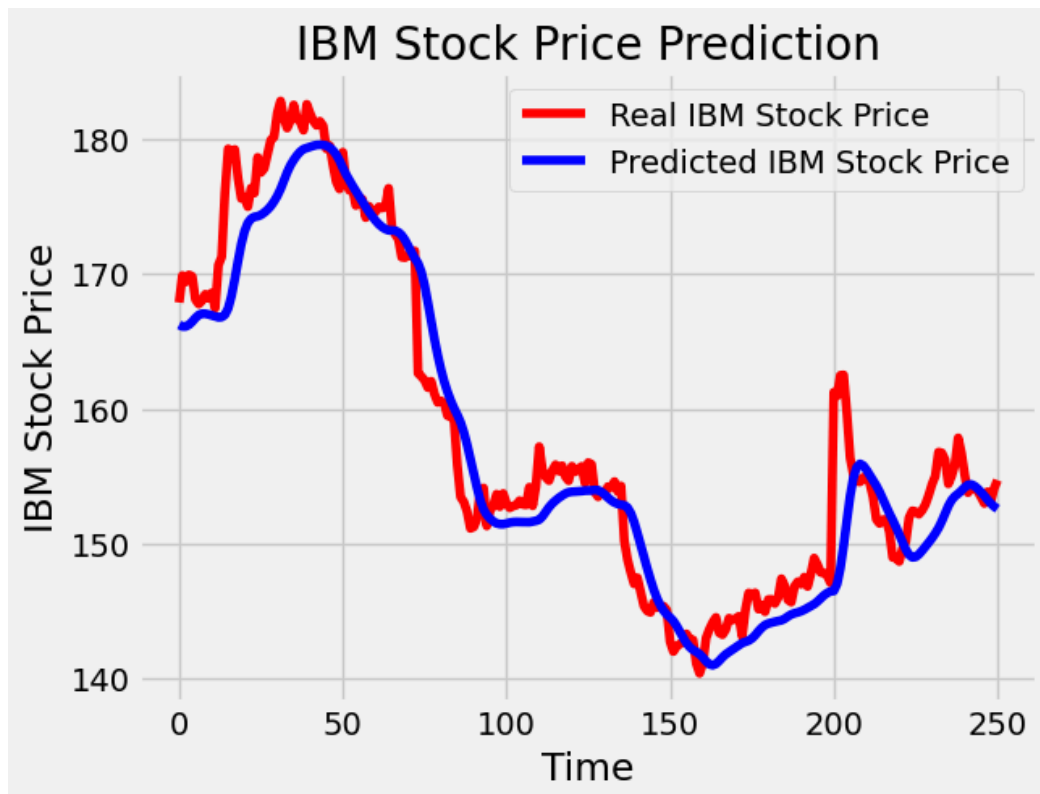
8/8 ————— 2s 178ms/step

```

```

# Visualizing the results for GRU
plot_predictions(test_set,GRU_predicted_stock_price)

```



```
# Evaluating GRU
return_rmse(test_set,GRU_predicted_stock_price)
```

The root mean squared error is 3.402293590013188.

✓ Sequence Generation

Here, I will generate a sequence using just initial 60 values instead of using last 60 values for every new prediction. **Due to doubts in various comments about predictions making use of test set values, I have decided to include sequence generation.** The above models make use of test set so it is using last 60 true values for predicting the new value(I will call it a benchmark). This is why the error is so low. Strong models can bring similar results like above models for sequences too but they require more than just data which has previous values. In case of stocks, we need to know the sentiments of the market, the movement of other stocks and a lot more. So, don't expect a remotely accurate plot. The error will be great and the best I can do is generate the trend similar to the test set.

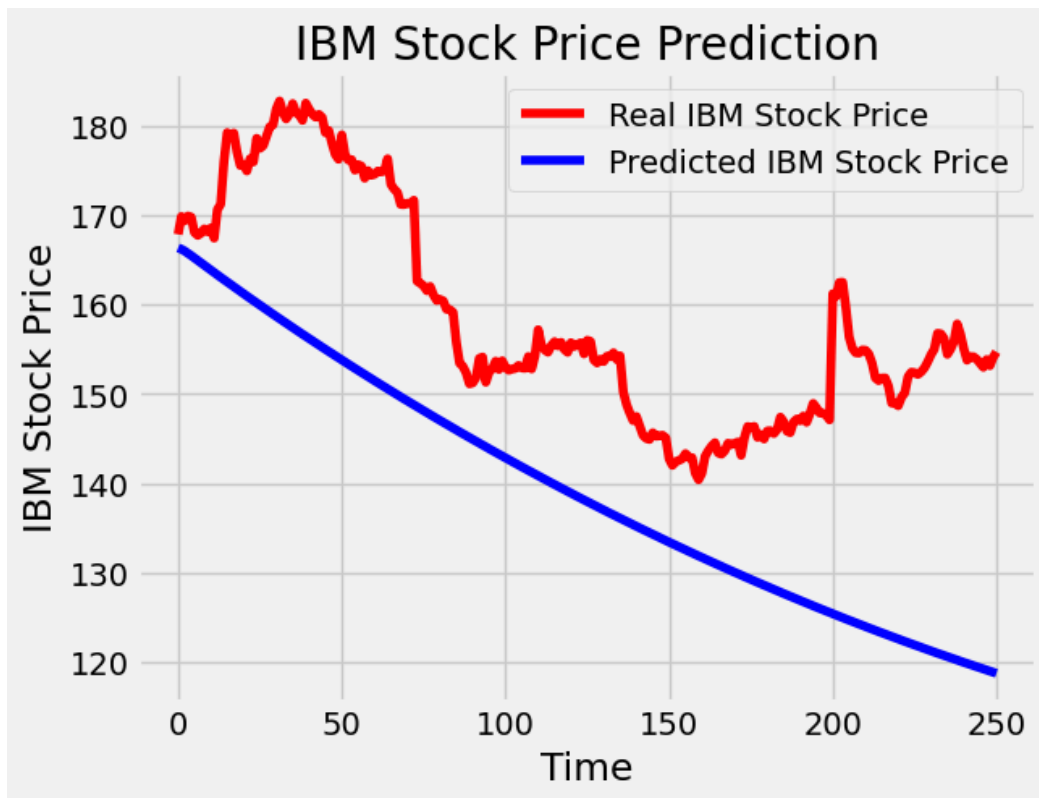
I will use GRU model for predictions. You can try this using LSTMs also. I have modified GRU model above to get the best sequence possible. I have run the model four times and two times I got error of around 8 to 9. The worst case had an error of around 11. Let's see what this iterations.

The GRU model in the previous versions is fine too. Just a little tweaking was required to get good sequences. **The main goal of this kernel is to show how to build RNN models. How you predict data and what kind of data you predict is up to you. I can't give you some 100 lines of code where you put the destination of training and test set and get world-class results. That's something you have to do yourself.**

```
# Preparing sequence data
initial_sequence = X_train[2708,:]
sequence = []
for i in range(251):
    new_prediction = regressorGRU.predict(initial_sequence.reshape(initial_sequence.shape[1],initial_
    initial_sequence = initial_sequence[1:]
    initial_sequence = np.append(initial_sequence,new_prediction,axis=0)
    sequence.append(new_prediction)
sequence = sc.inverse_transform(np.array(sequence).reshape(251,1))
```

```
1/1 _____ 0s 49ms/step
1/1 _____ 0s 49ms/step
1/1 _____ 0s 47ms/step
1/1 _____ 0s 54ms/step
1/1 _____ 0s 50ms/step
1/1 _____ 0s 52ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 88ms/step
1/1 _____ 0s 84ms/step
1/1 _____ 0s 84ms/step
1/1 _____ 0s 72ms/step
1/1 _____ 0s 67ms/step
1/1 _____ 0s 69ms/step
1/1 _____ 0s 74ms/step
1/1 _____ 0s 79ms/step
1/1 _____ 0s 72ms/step
1/1 _____ 0s 84ms/step
1/1 _____ 0s 80ms/step
1/1 _____ 0s 72ms/step
1/1 _____ 0s 81ms/step
1/1 _____ 0s 83ms/step
1/1 _____ 0s 76ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 49ms/step
1/1 _____ 0s 48ms/step
1/1 _____ 0s 72ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 58ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 59ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 50ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 48ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 49ms/step
1/1 _____ 0s 58ms/step
1/1 _____ 0s 67ms/step
1/1 _____ 0s 50ms/step
1/1 _____ 0s 52ms/step
1/1 _____ 0s 50ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 52ms/step
1/1 _____ 0s 51ms/step
1/1 _____ 0s 52ms/step
1/1 _____ 0s 54ms/step
1/1 _____ 0s 66ms/step
1/1 _____ 0s 52ms/step
1/1 _____ 0s 55ms/step
1/1 _____ 0s 52ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 50ms/step
1/1 _____ 0s 53ms/step
1/1 _____ 0s 50ms/step
```

```
# Visualizing the sequence
plot_predictions(test_set,sequence)
```



```
# Evaluating the sequence  
return_rmse(test_set, sequence)
```

The root mean squared error is 20.894216122449173.

So, GRU works better than LSTM in this case. Bidirectional LSTM is also a good way to make the model stronger. But this may vary for different data sets. **Applying both LSTM and GRU together gave even better results.**

I was going to cover text generation using LSTM but already an excellent kernel by [Shivam Bansal](#) on the mentioned topic exists. Link for that kernel here:

<https://www.kaggle.com/shivamb/beginners-guide-to-text-generation-using-lstms>

This is certainly not the end. Stay tuned for more stuff!